

GYUSEOK LEE

Ph.D. Student, School of Information Sciences
University of Illinois Urbana-Champaign
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RESEARCH INTERESTS

1. Recommender systems and personalization for real-world applications
2. Continual learning and large language models for user modeling

EDUCATION

Ph.D., Information Sciences, University of Illinois Urbana-Champaign Aug 2025 - Present
Advisor: [Prof. Jingrui He](#)
Cumulative GPA: 3.83 / 4.00

M.S., Artificial Intelligence, Pohang University of Science and Technology Feb 2022 - Feb 2024
Advisor: [Prof. Hwanjo Yu](#)
Cumulative GPA: 3.77 / 4.30

Thesis: Collaborative Knowledge Distillation for Continual Learning in Recommender System

B.S., Computer Science, Mathematics and Statistics, Handong Global University Mar 2015 - Feb 2022
Advisors: [Prof. InJung Kim](#) and Prof. Heonjoo Kim
Cumulative GPA: 4.01 / 4.50 & Major GPA: 4.22 / 4.50 (Magna Cum Laude)
Includes 23 months of mandatory military service in the Republic of Korea Air Force

INDUSTRY EXPERIENCE

Applied Scientist Intern, Amazon Jun 2026 - Sep 2026
Cold-start problem in large-scale recommender systems
Manager: [Shreya Chakrabarti](#)

Medical AI Research Intern, AITRICS Dec 2021 - Feb 2022
Machine learning for clinical prediction and pathology imaging
1. Two-stage CNN and XGBoost pipeline for thyroid cancer prediction
2. Self-supervised learning for pathology image classification with limited annotations

ACADEMIC EXPERIENCE

Visiting Researcher, Korea University Apr 2025 - Aug 2025
LLM-generated intents for session-based recommendation [SIGIR 2026]
Host: [Prof. SeongKu Kang](#)

Researcher, Pohang University of Science and Technology Mar 2024 - Aug 2025
Continual sequential recommendation under non-stationary data streams [WSDM 2026]
Host: [Prof. Hwanjo Yu](#)

Visiting Researcher, University of Virginia

Collaborative diffusion models for recommender systems [WWW 2025]

Host: [Prof. Jundong Li](#)

Sep 2023 - Feb 2024

Charlottesville, VA

Visiting Researcher, Carnegie Mellon University

IITP AI Intensive Program, sponsored by the Korean Government

Aug 2022 - Feb 2023

Pittsburgh, PA

1. Visual clustering for personalized multimodal fashion recommendation (project lead)
2. Multi-agent reinforcement learning for taxi repositioning with attention networks (project lead)
3. Masked language modeling for question answering and question generation
4. Generative model for automatic paper assessment and review scoring

Graduate Research Assistant, Pohang University of Science and Technology

Knowledge distillation for continually updated recommender systems [KDD 2024]

Feb 2022 - Feb 2024

Pohang, South Korea

Advisor: [Prof. Hwanjo Yu](#)

Undergraduate Research Intern, Handong Global University

Partial learning algorithms for on-device deep neural network training

Feb 2021 - Dec 2021

Pohang, South Korea

Supported by ETRI (Electronics and Telecommunications Research Institute)

Advisor: [Prof. InJung Kim](#)

PUBLICATIONS

* Equal contribution.

First-Authored (6)

1. SCoRD: Semantic-Assisted Continual Retriever-Reranker Distillation for LLM-Based Recommendation
[Seunghyun Baek*](#), [Gyuseok Lee*](#), [Seunghan Lee](#), [Wonbin Kweon](#), [Dong Wang](#), [SeongKu Kang](#)
CIKM 2026 | ACM International Conference on Information and Knowledge Management
2. Uncertainty-Aware Variational Reward Factorization via Probabilistic Preference Bases for LLM Personalization
[Gyuseok Lee](#), [Wonbin Kweon](#), [Zhenrui Yue](#), [SeongKu Kang](#), [Jiawei Han](#), [Dong Wang](#)
COLM 2026 | Conference on Language Modeling
3. SPRINT: Scalable and Predictive Intent Refinement for LLM-Enhanced Session-Based Recommendation
[Gyuseok Lee](#), [Wonbin Kweon](#), [Zhenrui Yue](#), [Yaokun Liu](#), [Yifan Liu](#), [Susik Yoon](#), [Dong Wang](#), [SeongKu Kang](#)
SIGIR 2026 | ACM SIGIR Conference on Research and Development in Information Retrieval
4. Capturing User Interests from Data Streams for Continual Sequential Recommendation
[Gyuseok Lee](#), [Hyunsik Yoo](#), [Junyoung Hwang](#), [SeongKu Kang](#), [Hwanjo Yu](#)
WSDM 2026 | ACM International Conference on Web Search and Data Mining
5. Collaborative Diffusion Model for Recommender System
[Gyuseok Lee](#), [Yaochen Zhu](#), [Hwanjo Yu](#), [Yao Zhou](#), [Jundong Li](#)
WWW 2025 | ACM The Web Conference, Short Paper
6. Continual Collaborative Distillation for Recommender System
[Gyuseok Lee*](#), [SeongKu Kang*](#), [Wonbin Kweon](#), [Hwanjo Yu](#)
KDD 2024 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

Co-Authored (2)

7. Topology-Aware Tokenization for Generative Recommendation
[Yaokun Liu](#), [Yifan Liu](#), [Zhenrui Yue](#), [Gyuseok Lee](#), [Zelin Li](#), [Ruichen Yao](#), [Dong Wang](#)

8. Harmonic Dataset Distillation for Time Series Forecasting
Seungha Hong, Sanghwan Jang, Wonbin Kweon, Suyeon Kim, Gyuseok Lee, Hwanjo Yu
AAAI 2026 | AAAI Conference on Artificial Intelligence

Under Review (1)

9. ContextRank: Enhancing Scientific Document Retrieval with Context-Enriched Relations
Wonbin Kweon, Xueqiang Xu, Runchu Tian, Chih-Hsuan Wei, ZhiZheng Wang, Hyunsik Yoo, Gyuseok Lee, SeongKu Kang, Zhiyong Lu, Jiawei Han, Hwanjo Yu
EMNLP 2026 | Conference on Empirical Methods in Natural Language Processing

AWARDS

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| 1. NSF Travel Award (\$1,500), ACM WSDM | Feb 2026 |
| 2. Facilitating Learning Excellence Award, University of Illinois Urbana-Champaign | Dec 2025 |

SCHOLARSHIPS

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| 1. Foundation Scholarship (\$2,000), OTEFE Foundation, USA | Aug 2026 |
| 2. Foundation Scholarship (\$1,000), OTEFE Foundation, USA | Aug 2025 |
| 3. Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea | Nov 2021 |
| 4. Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea | Mar 2021 |
| 5. Academic Excellence Scholarship, Handong Global University, South Korea | Mar 2021 |
| 6. Academic Excellence Scholarship, Handong Global University, South Korea | Sep 2020 |
| 7. Academic Excellence Scholarship, Handong Global University, South Korea | Mar 2020 |

PROFESSIONAL SERVICE

Reviewer for all entries.

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|---|------------|
| 1. ACM Transactions on Information Systems (TOIS) | 2025, 2026 |
| 2. Conference on Language Modeling (COLM) | 2026 |
| 3. ACM SIGIR Conference (SIGIR) | 2026 |
| 4. ACM The Web Conference (WWW) | 2025 |

TEACHING

Teaching assistant unless otherwise noted.

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|---|-------------------|
| 1. Foundations of Data Curation, University of Illinois Urbana-Champaign | Fall 2026 |
| 2. Reading and Writing Data, University of Illinois Urbana-Champaign | Spring 2026 |
| 3. Data Mining, University of Illinois Urbana-Champaign | Fall 2025 |
| 4. Artificial Intelligence, SK industry training program (instructor) | Summer 2024, 2025 |
| 5. Object-Oriented Programming, Pohang University of Science and Technology | Spring 2023 |
| 6. Big Data Analysis with R, Handong Global University | Fall 2021 |
| 7. Mathematical Statistics, Handong Global University | Summer 2021 |
| 8. Statistical Methodology, Handong Global University | Spring 2021 |
| 9. Python Programming, Handong Global University and KMOOC | Spring, Fall 2020 |